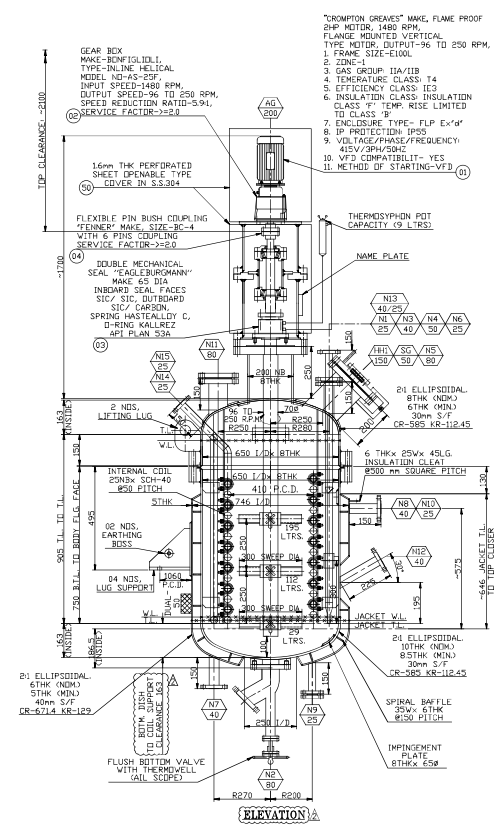
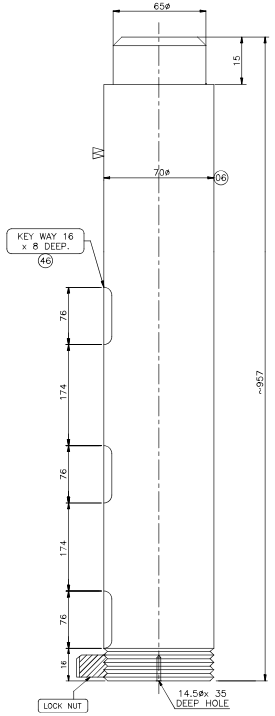
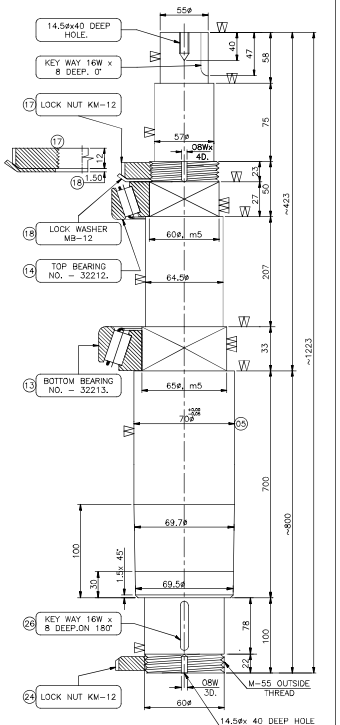
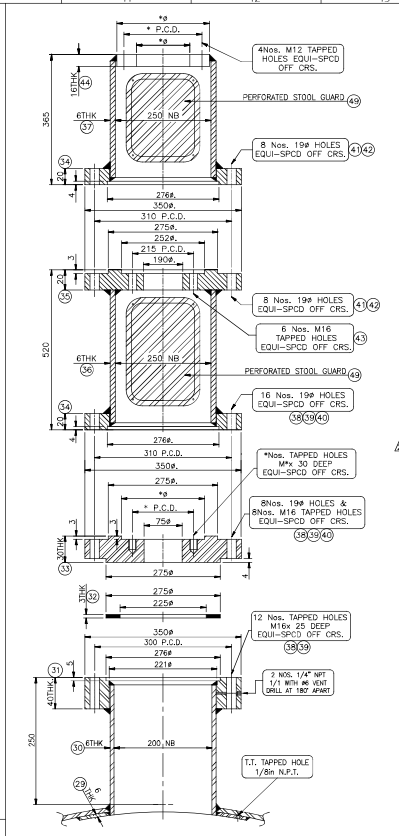




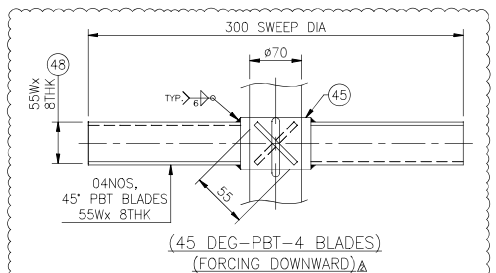
BOTTOM SHAFT DETAIL.



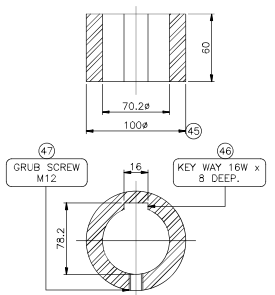
TOP SHAFT DETAIL.



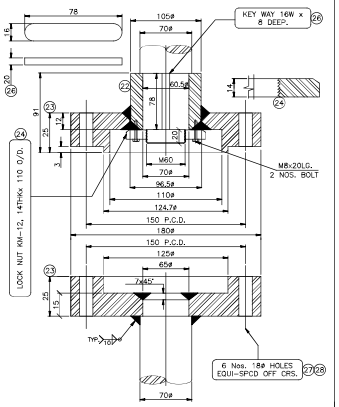
STOOL ASSLY. WITH TOP PAD DTL.
*DIMENSION TO SUIT 70ø MECHNICAL SEAL



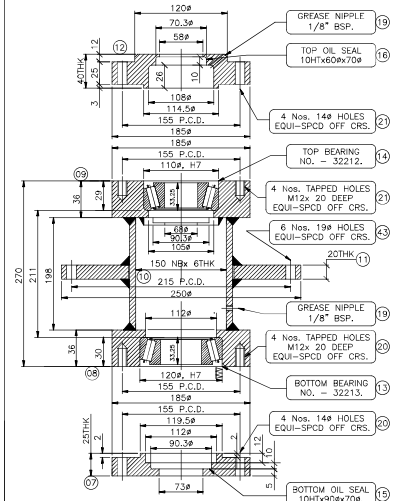
(45 DEG-PBT-4 BLADES) (FORCING DOWNWARD)



BUSH DETAIL.



INTERNAL RIGID COUPLING DETAIL.



BEARING HOUSING & COVER DETAIL.

DESIGN DATA

DESIGN CODE:- ASME SEC-VIII, DIV-1, LATEST EDITION 2021

		SHELL SIDE	JACKET SIDE	INTERNAL COIL
OPERATING PRESSURE, (INT./EXT.)	Kg/Cm2.(g).	ATM	3/-	-
OPERATING TEMPERATURE, (MIN./ NOR. / MAX.)	°C	50	35	-
SERVICE FLUID		-	BRINE AND STEAM	CHILLED WATER/ COOLING WATER/ CHILLED BRINE
SERVICE FLUID CATEGORY		TOXIC	-	-
DESIGN PRESSURE, (INT./EXT.)	Kg/Cm2.(g).	6/F.V.	6/-	6/-
DESIGN TEMPERATURE (MIN/MAX)	°C	150	(-)20/150	150
MAXIMUM ALLOWABLE WORKING PRESSURE (MAWP)	Kg/Cm2.(g).	6	6	6
MINIMUM DESIGN METAL TEMPERATURE (MDMT)	°C	(-)29	(-) 29	(-) 29
DENSITY OF FLUID	Kg/M3.	1800	1050	1050
CORROSION ALLOWANCE.	mm	NIL	NIL	NA
WELD JOINT EFFICIENCY. (SHELL/DISH)		0.85/1.0	0.85/1.0	1
RADIOGRAPHY. (SHELL/DISH)		10% SPOT/ FULL	10% SPOT/ FULL	FULL
TEST PRESSURE	Kg/Cm2.(g).	7.8	7.8	7.8
HEAT TRANSFER AREA	~M^2	~~	~1.88	~1.89
CAPACITY (GROSS/WORK)	KL	~0.33/ ~0.26		
INSULATION TYPE/ THICNESS/ MATERIAL	mm	DUM./ 50/ --		
EMPTY WEIGHT.	KG	~1760		
OPERATING WEIGHT.	KG	~2020		
HYDRO TEST WEIGHT.	KG	~2090		
WIND DESIGN CODE		IS-875 (PART-3)		
BASIC WIND SPEED		44 m/sec.		
SEISMIC DESIGN CODE		IS-1893 RSM (PART 2 & 4), (ZONE-II)		

NOTES.

(01) ALL DIMENSIONS ARE IN mm UNLESS OTHERWISE SPECIFIED.
(02) LANTER STOOL & BRG HOUSING DIMENSIONS ARE INDICATIVE ONLY & MAY VERY WITH ACTUAL ASSEMBLY HOWEVER THE SAME WILL BE INCORPORATED IN AS BUILT FABRICATION DRG.

Add following details:
Design Code : EEUA
Degree of Agitation: 4.16
Mixing Cycle: --
Batch Cycle time: --
Reaction: --
Noise Level: <= 85 dB(A) @ 1 meter distance
Critical Speed: 634.23rpm
Dynamic Balancing: ISO 21940-11, Gr. 6.3
Shaft Power (BKW): 1.8HP

1) REACTION/LOAD AT SUPPORT- △
A)WIND LOAD-861.86N
B)SEISMIC LOAD IN OPERATING CONDITION-4092.56N
C)SEISMIC LOAD IN EMPTY CONDITION-3273.84N
D)SEISMIC LOAD IN DYNAMIC CONDITION-6138.84N
2) VESSEL CG -
A)CG IN OPERATING CONDITION-601.23MM
B)CG IN EMPTY CONDITION-683.88MM

REV.	DATE.	DESCRIPTION.	DRAWN.	CHKD.
R2.	03.05.2023	COMMENTS INCORPORATED & SUBMITTED FOR APPROVAL	W.M.	D.S.
R1.	20.03.2023	COMMENTS INCORPORATED & SUBMITTED FOR APPROVAL	W.M.	D.S.
R0.	26.12.2022	FOR APPROVAL.	W.M.	D.S.

NO.	DESCRIPTION.	QTY.	MATERIAL.	SIZE.	REMK.
50)	PERFORATED OPENABLE TYPE COVER	2	S.S.304.	1.6THKx TO SUIT.	
49)	PERFORATED STOOL GUARD	4	S.S.304.	3THKx TO SUIT.	
48)	PITCH BLADE TURBINE.	12	SA.240 GR.316L.	8THKx 55W.	
47)	GRUB SCREW.	3	S.S.316.	M12x 50 SUIT LG.	
46)	GIB-HD KEY.	3	SA.240 GR.316L.	AS PER DETAIL	
45)	BUSH FOR AGITATOR.	3	SA.479 GR.F.316L.	65.2ø x 95ø x 60HT.	
44)	BASE PL. FOR MOTOR & G.B.	1	IS.2062 GR.B.	16THKx TO SUIT LG.	
43)	HEX-HD BOLTS (BRG. SUPP.)	6	IS.1367 CL.4.6.	M16x 40LG.(MIN.)	
42)	HEX-HD NUTS.	8	IS.1367 CL.4.0.	M16.	
41)	HEX-HD BOLTS.	8	IS.1367 CL.4.6.	M16x 60LG.(MIN.)	
40)	HEX-HD BOLTS (PAD).	8	IS.1367 CL.4.6.	M16x 45LG.(MIN.)	
39)	HEX-HD BOLTS (PAD).	8	IS.1367 CL.4.6.	M16x 65LG.(MIN.)	
37)	TOP STOOL PIPE.	1	IS.2062 GR.B.	6THKx 250NB.	
36)	BOTTOM STOOL PIPE.	1	IS.2062 GR.B.	6THKx 250NB.	
35)	FLANGE RING FOR STOOL PIPE.	1	IS.2062 GR.B.	20THKx 185ø x 350ø.	
34)	FLANGE FOR STOOL PIPE.	2	IS.2062 GR.B.	20THKx 276ø x 350ø.	
33)	SAND WITCH PLATE	1	SA.240 GR.316L.	30THKx 75ø x 350ø.	
32)	GASKET.	1	SOLID PTFE.	3THKx 225 I/Dx 275 O/D.	
31)	FLANGE.	1	SA.240 GR.316L.	40THKx 221 I/Dx 350 O/D.	
30)	NECK.	1	SA.240 GR.316L.	200NBx 8THKx 250LG.	FABRICATED
29)	R.F. PAD	1	SA.240 GR.316L.	6THKx 221 I/Dx 325 O/D.	
28)	HEX-HD NUTS.	6	S.S.316.	M16.	
27)	HEX-HD BOLTS.	6	S.S.316.	M16x 70LG.(MIN.)	
26)	PARALLEL KEY FOR R.C.	1	SA.240 GR.316L.	20THKx 16Wx 81LG.	
24)	LOCK NUT FOR RIGID CPLG.	1	S.S.316L.	KM-12.	
23)	FLG. FOR INTER. RIGID CPLG.	2	SA.240 GR.316L.	25THKx 65ø/105ø x 180ø	
22)	BUSH FOR INTER. RIGID CPLG.	1	S.S.316L.	60.5ø x 105ø x 78HI.	
21)	HEX-HD BOLTS.	4	IS.1367 CL.4.6.	M12x 30LG.	
20)	HEX-HD BOLTS.	4	IS.1367 CL.4.6.	M12x 40LG.	
19)	GREASE NIPPLE.	2	M.S.	1/8" BSP.	
18)	LOCK WASHER.	1	STD.	MB-12.	
17)	LOCK NUT.	1	STD.	KM-12.	
16)	TOP OIL SEAL.	1	RUBBER.	60ø x 70ø x 10HT.	
15)	BOTTOM OIL SEAL.	1	RUBBER.	70ø x 90ø x 10HT.	
14)	TOP TAPPER ROLLER BEARING.	1	STD,SKF.	BRG. NO.-32212.	
13)	BOTTOM TAP. ROLLER BEARING.	1	STD,SKF.	BRG. NO.-32213.	
12)	TOP BEARING COVER.	1	IS.2062 GR.B.	40THKx 58ø x 185ø.	
11)	BEARING HOUSING SUPPORT.	1	IS.2062 GR.B.	20THKx 171ø x 250ø	
10)	PIPE FOR BEARING HSG.	1	IS.2062 GR.B.	6THKx 150NBx 198HT.	
09)	TOP BRG. HSG. PLATE.	1	IS.2062 GR.B.	36THKx 105ø x 185ø.	
08)	BOTTOM BRG. HSG. PLATE.	1	IS.2062 GR.B.	36THKx 112ø x 185ø.	
07)	BOTTOM BEARING COVER.	1	IS.2062 GR.B.	25THKx 73ø x 185ø.	
06)	BOTTOM SHAFT.	1	SA.479 GR.F.316L.	70ø x ~957HT.	
05)	TOP SHAFT.	1	SA.479 GR.F.316L.	70ø x ~1223HT.	
04)	FLEXIBLE COUPLING.	1	"FENNER" MAKE.	PIN BUSH COUPLING SIZE-BC-4 WITH 6 PINS COUPLING SERVICE FACTOR->=2.0	
03)	MECHANICAL SEAL.	1	"EAGLEBURGMANN" MAKE	DOUBLE MECHANICAL SEAL 65 DIA INBOARD SEAL FACES SIC/ SIC, OUTBOARD SIC/CARBON, SPRING HASTEALLOY C, O-RING KALLREZ API PLAN 53A	
02)	GEAR BOX.	1	"BONFIGLIOLI" MAKE	TYPE-INLINE HELICAL MODEL NO-AS-25F, INPUT SPEED-1480 RPM, OUTPUT SPEED-96 TO 250 RPM, SPEED REDUCTION RATIO-5.9:1, SERVICE FACTOR->=2.0	
01)	MOTOR.	1	"CROMPTON GREAVES" MAKE	2HP MOTOR, 1480 RPM, FLAME PROOF FLANGE MOUNTED VERTICAL TYPE MOTOR, OUTPUT-96 TO 250 RPM, 1. FRAME SIZE-E100L 2. ZONE-1 3. GAS GROUP: IIA/IIB 4. TEMERATURE CLASS: T4 5. EFFICIENCY CLASS: IE3 6. INSULATION CLASS: INSULATION CLASS 'F' TEMP. RISE LIMITED TO CLASS 'B'. 7. ENCLOSURE TYPE-FLP Ex'd' 8. IP PROTECTION: IP55 9. VOLTAGE/PHASE/FREQUENCY:415V/3PH/50HZ 10. VFD COMPATIBILITY-YES 11. METHOD OF STARTING -VFD	

AEROLITE ENGINEERS PVT. LTD.
PLOT. NO. : N-64, MIDC, TARAPUR, BOISAR DIST., PALGHAR- 401506.

CLIENT.	:M/S. AARTI INDUSTRIES LIMITED	DRAWN. W.M.	26.12.2022
PROJECT	:PILOT PLANT, JHAGADIA, GUJARAT (50)	CHKD. D.S.	26.12.2022
INSP BY	:M/S. CLIENT/TPI	APPRD.	
REF DRG NO.	:425124-MMD-00-XX-DR-PE-0005 REV:C02	CAD NO.	22041C
P.O. NO.	:4550406897 DATE. 23.12.2022	SCALE: AS NOTED	SHEET 3 OF 3
TAG NO	:11R0101 QTY. 1NO.	DRG NO: AEPL-221223C	
TITTLE.	:REACTOR (11R0101)	REV.2 △	